
6 Faults and fault finding

6.1 The fault display

The PPC6000 series carry out a number of internal and external checks during operation. If a fault is found, a fault number (displayed with a text description) is used to identify the type of problem. A list of fault numbers may be found later in this chapter.

Many of the faults detected and displayed by the control will cause a non-volatile lockout of the burner, that is to say that removing the power to the control will not cancel a fault. Other faults will be displayed as a warning but operation of the burner will not be affected.

Each fault number is prefixed on the display by a letter as follows:

- F** This means that a fault has occurred and is still present.
The fault may be internal or external to the control.
- C** This means that a fault did occur but has now cleared.
- L** This means a programmed limit has been exceeded.

For example:

217 PSI Actual AUTO
F23:Drive 3 Fault

Many of the faults detected and displayed by the control will cause a non-volatile lockout of the burner. Other faults will be displayed as a warning but operation of the burner will not be affected. A complete list of faults may be found later in this chapter.

A fault history is available through the Engineers Key, the standard fault history is lost if the power is removed from the unit but as an option the fault history may be displayed with a date and time stamp, and retained in the unit memory in the event of a power loss to the control. For further information see section 6.6.1.

NOTE: LED on PPC6000: STATUS LED: Flashing LED indicates the control is OK. OFF = possible *open fuse 2* (F32 see Section 2.6.1) Also, OFF = possible *fatal processor fault*.

6.2 What to do when a fault occurs

If faults or limits are present and either the alarm or limit relay is closed, press the **MUTE/RESET** key to mute the alarm (open the relay).

Take note of the fault message and use the Fault Listing to identify why the fault occurred.

If the faults, which cause a non-volatile lockout, are still present when the alarm is muted, the alarm will operate again when the alarm clears, to indicate to the operator that the control may be restarted.

Once all faults have been diagnosed, rectified and cleared, press the **MUTE/RESET** key for approximately three seconds to remove the fault messages and re-start the burner.



The function of the **MUTE/RESET** key may also be achieved via ComFire or using the MUTE/RESET input. See option parameter 1.2.

6.3 Non-volatile lockout

If power is removed from the control while a fault is still present, the fault will be stored in non-volatile memory. When power is restored to the control, the fault number will still be present and it will be necessary to remove the fault before restarting the burner.

6.4 Fault subsets

As an aid to fault finding, most faults also have a *fault subset* that gives additional information about the type of fault or what the burner was doing when the fault occurred. Where applicable, the subsets are given in the fault listing in section 6.5

6.5 Fault listing

Fault Number	Cause	Description
F01	External Alarm Fault 1	One of the low voltage alarm / lockout inputs is causing an alarm.
F02	External Alarm Fault 2	The fault may be prefixed by either an 'L' or an 'F' and may or may not shutdown / lockout the burner.
F03	External Alarm Fault 3	See Option parameters 19.1 to 19.4.
F04	External Alarm Fault 4	<i>Subset: burner status.</i>
F18	Not enough setpoints entered	A fuel/air ratio profile has been selected which does not have at least four setpoints commissioned. • Use commission ratio mode to enter more setpoints. <i>Subset: fuel code</i>
F19	Circuit board voltage reference fault	The circuit board has an incorrect on-board reference level. • Ensure ALL analog inputs (4-20mA, 0 – 5v) are in the range 0 to 5.0 volts. It is critical that none of the inputs are higher than 5.0 volts. • If the problem persists even when all analog inputs are disconnected, check EK37 and contact supplier. <i>Subset: +1 out of range, +2 zero check failed, +4 span check failed.</i> See EK37 for more details.



Fault Number	Cause	Description														
F20	Drive 0 position fault	A drive-positioning fault has occurred, which will cause a non-volatile lockout of the burner. This fault has occurred for one of the following reasons: 1. During close position prove, when a drive stops at a position that is outside the close set position by more than five degrees. 2. During purge position prove, when a drive stops at a position that is outside the purge set position by more than five degrees. 3. During pre-purge, ignition or post-purge, when a drive moves from its setpoint. 4. During modulation, when a drive is not at its correct setpoint as defined by the commissioned fuel/air ratio for the selected profile. A drive is defined as having moved from its setpoint if its positional error is more than 1° for 15s, or more than 5° for 1s. For positional errors between 1° and 5°, the detection time is variable between 15s and 1s. Note: Only the selected drives are checked, i.e. used on the current fuel/ air profile. The other drives are ignored. <i>Subset: burner status.</i> 000 - 016 = Position fault at the status number 032 – 048 = CAN communications error. Display shows ERR1 under drive name. > 064 = Internal servo fault., Display shows ERR2 - ERR7 under the drive name. Note: A message ERR1, ERR2, ERR3 on the display indicates a communications failure with a given servo-motor. This could be caused by a wiring error or an incorrectly selected serial number. Verify selections in Option parameter 03.x.														
F21	Drive 1 position fault															
F22	Drive 2 position fault															
F23	Drive 3 position fault															
F24	Drive 4 position fault															
F25	Drive 5 position fault															
F26	Drive 6 position fault															
F27	Drive 7 position fault															
F28	Drive 8 position fault															
F29	Drive 9 position fault															
F32	Safety input fault	One or more of the fail-safe low voltage inputs is registering a fault. <i>Subset: Combination of failed inputs.</i> <table><tr><th>Subset #</th><th>Input I</th></tr><tr><td>0</td><td>N/A</td></tr><tr><td>1</td><td>1</td></tr><tr><td>2</td><td>2</td></tr><tr><td>4</td><td>3</td></tr><tr><td>8</td><td>4</td></tr><tr><td>128</td><td>Air pressure input</td></tr></table>	Subset #	Input I	0	N/A	1	1	2	2	4	3	8	4	128	Air pressure input
Subset #	Input I															
0	N/A															
1	1															
2	2															
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8	4															
128	Air pressure input															

Fault Number	Cause	Description
F33	Burner input fault	One or more of the high voltage (profile select) inputs is registering a fault. <i>Subset: Failure mode.</i>
F34	Primary relay fault	One or more of the internal relays is not responding correctly. <i>Subset : 1 - 10 = Failed relay number</i>
F35	ADC fault	One of the internal checks on the analog to digital converter has failed. <i>Subset: Failure mode.</i>
F36	Reset fault	This fault occurs if the control receives more than 5 'false' restart requests within a 15 minute period. A false restart request occurs then the fault mute key / input is held for more then three seconds when there is no fault present. This is to protect against a failure in an external device continually re-starting the control, which could cause un-attended burner restarts in the event of a genuine burner problem. This is important since the fault mute function can come from anywhere via a digital input or serial communications. To clear the fault, either enter a commission mode, or cycle the power to the control. The subset in not applicable for this fault
F37	RAM test fault	The main memory in the control has malfunctioned. • Interrupt power to the control. • This could be due to high levels of electrical interference getting into the product. Check all cables are correctly screened, and screens are terminated correctly. Ensure the mains supply is not excessively noisy. • If this fault persists, the control must be returned to the supplier. <i>Subset: Failed Page.</i>

Fault Number	Cause	Description
F38	Program memory CRC fault	The program memory in the control has been corrupted. • Interrupt power to the control. • This could be due to high levels of electrical interference getting into the product. Check all cables are correctly screened, and screens are terminated correctly. Ensure the mains supply is not excessively noisy. • If this fault persists, the control must be returned to the supplier. <i>Subset: Failed Page.</i>
F39	Profile table CRC fault	The profile table memory in the control has been corrupted. • Interrupt power to the control. • Erase the system using option parameter 45.1 = 5 and re commission (or restore with option parameter 45.2). • This could be due to high levels of electrical interference getting into the product. Check all cables are correctly screened, and screens are terminated correctly. Ensure the mains supply is not excessively noisy. • If this fault persists, the control must be returned to the supplier. <i>Subset: Failed Page.</i>
F40	Spare	N/A
F41	Boiler safety limit exceeded	The boiler's measured value has exceeded the pressure/temperature safety limit. <i>Subset: 4 - safety limit exceeded</i>
F42	<i>Reserved</i>	
F43	<i>Reserved</i>	
F44	Burner not OFF.	This fault will lock the burner out if the AUTO input (PA10 – PA11) is made when the control thinks the burner should not be firing. It will also occur if the AUTO input stays ON for three seconds after this control has opened the 'controlled shutdown relay' – i.e. tried to turn the burner off. <i>Subset: burner status.</i>
F45	<i>Reserved</i>	

Fault Number	Cause	Description
F46	EEPROM memory CRC fault	The EEPROM memory in the control has been corrupted. This memory is used to store the option parameters, - Interrupt power to the control. - Erase the system using option parameter 45.1 = 5 and re commission (or restore with option parameter 45.2). - This could be due to high levels of electrical interference getting into the product. Check all cables are correctly screened, and screens are terminated correctly. Ensure the mains supply is not excessively noisy. - If this fault persists, the control must be returned to the supplier. <i>Subset: Failed Page.</i>
F50	Oxygen probe heater fault	The probe heater has failed to heat to the correct temperature after 30 minutes of system power up. - Is the probe heater wiring correct? - Is the probe cell thermocouple wiring correct? Once the fault is rectified, it will be necessary to interrupt the power to the oxygen probe interface to make it attempt to heat the probe once more. <i>Subset: O2 probe status.</i>
L52	Oxygen low limit alarm	The oxygen level measured value is below the oxygen setpoint low alarm value for the current profile. <i>Subset: 0 = Limit violation, 1 = Probe Failed.</i>
L53	Oxygen high limit alarm	The oxygen level measured value has exceeded the oxygen setpoint high alarm value for the current profile. <i>Subset: 0 = Limit violation, 1 = Probe Failed.</i>
F54	Oxygen probes mismatch fault	This fault occurs when two oxygen probes are used to give fail-safe oxygen monitoring. If this fault occurs, check and calibrate both oxygen probes. It may be necessary to increase option parameter 42.4 or 42.5 (within safe limits) if the oxygen probes are in different parts of the flue. Subset: 1 = Oxygen values don't match, 2 = Flue temperatures don't match 3 = Neither flue temperature or oxygen levels match 255 = Second oxygen probe is faulty / not ready. See EK 76.

Fault Number	Cause	Description
L55	Trim limit alarm	The trim drive has reached the allowed maximum deviation limit. • Change trim limit. • Re-commission fuel / air ratio. <i>Subset: Not applicable.</i>
F57	Auto trim commissioning fault	The measured oxygen level exceeded 15.0% during auto trim commissioning. The burner is shutdown. <i>Subset: Last auto trim commission setpoint (+32 if adding air).</i>
L58	Flue temperature low alarm value exceeded	The measured flue temperature is below the low alarm value for the current profile, or the flue thermocouple is faulty. <i>Subset: Not applicable.</i>
L59	Flue temperature high alarm value exceeded	The measured flue temperature has exceeded the high alarm value for the current profile. <i>Subset: Not applicable.</i>
F60	Spare	Currently unassigned
F61	Combustion air pressure fault	Applicable if combustion air pressure monitoring is used (using a 4-20mA pressure sensor). This fault will lock out the burner if the measured combustion air pressure does not match that expected for the current firing rate for six (6) seconds. <i>Subset is not used (will be zero).</i>
F62	Spare	Currently unassigned
F63	Option parameters uploaded	The option parameters have been uploaded via serial communications. Check all values are correct and match the application, then set option parameter 45.0 to 0. <i>Subset : Not applicable.</i>



Fault Number	Cause	Description
F64	Profile Invalid	This fault means that the control can't fire the currently selected profile because it isn't sure it still matches the burner / site configuration. It could lead to a hazardous situation. There are three possible causes to this fault: 1 – A drive has been commissioned in this profile but has now been deselected. If a profile is commissioned with two air dampers but then the second air damper is de-selected with option parameter 4.0 to 4.9, the profile is no longer valid. <i>Subset: Drive number (0 to 9)</i> 2 – Servo / Drive changed. If the system is commissioned then later one of the servos is changed for a new one (i.e. different serial number), any profiles that use that servo are now considered invalid. This is to force the engineer to ensure the new servo is mechanically connected the same way as the original one. <i>Subset: Invalid profile + 100 (101 to 104).</i> 3 – The selected profile has been uploaded from a PC, but has not been verified on this burner. <i>Subset: Invalid profile + 100 (101 to 104).</i> In all cases the F74 can be cleared by switching to another (good) profile, or by re-commissioning the profile in 'commission ratio' mode, making sure all points up to and including high fire are acknowledged using the 'NEXT' key (or 'ENTER' if the points are adjusted).
F65	Power-up Lockout	The control has locked out on power-up. This will normally be because option parameter 1.0 is set to 1. <i>Subset:</i> 1 – See option parameter 1.0. 254 – Serial EEPROM write failure. 255 – NV Lockout verification failed.
F66	Spare	Currently unassigned
F67	Secondary relay fault	A secondary fault has occurred with the non-volatile lockout relay. A Self check is performed once an hour and two consecutive failures will cause the fault to appear and the burner to lock-out immediately. • Contact supplier <i>Subset: Failed relay number.</i>

Fault Number	Cause	Description
F68	Secondary program memory checksum fault	A fault has occurred with the program memory in the control. A Self check is performed once an hour and two consecutive failures will cause the fault to appear and the burner to lock-out immediately. Contact supplier <i>Subset: Not applicable.</i>
F69	Secondary watchdog fault	A fault has occurred with the CPU watchdog. A Self check is performed once an hour and two consecutive failures will cause the fault to appear and the burner to lock-out immediately. Contact supplier <i>Subset:</i> <i>1 = Late test failed.</i> <i>2 = Early test failed.</i>
F70 – F79	User Faults	These fault numbers are generated by the user programmable section of the control and will vary with the application.

6.6 The engineer's key

By selecting the engineer's key it is possible to read the values of internal system variables and external input and output states. It is also possible to see the values of *fault subsets*, in order to obtain more detailed information about a fault that has occurred.

When using the engineer's key it is not possible to change any parameters. Therefore, it is not possible to affect the operation of the burner.

After pressing the Engineer's Key:

- The top line of the display will show an EK (data register) number and the value of the register.
- The bottom line of the display will show the EK description.

For example:

EK025= 2
 Burner Status

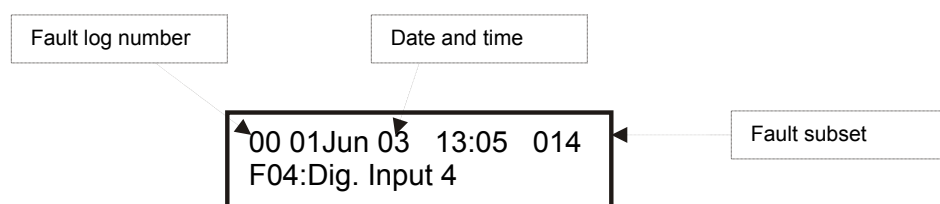
- If there is a fault present, this will be shown on the right hand side of the top line in the format Fnn,sss where nn = fault number, and sss = fault subset value. Note, this fault information will only be shown when viewing EK numbers 1 to 100.
- Use the UP/DOWN keys to change the EK number and view its corresponding data.
- Press the EK to index back to the normal display. The control will revert automatically to the normal run display if a key has not been pressed for one minute.

6.6.1 Fault history

The PPC6000 series stores information on faults. The memory in the unit will store the last 128 faults and associated status data and as an option a date and time stamp. The 100 most recent faults are available via the display unit and all 128 are available through ComFire. The Engineers Key gives access to the fault history.

To view the fault history scroll up through the EK's numbers until EK200 then by scrolling once more the most recently occurring fault is displayed, indicated by 00 in the top left corner of the display.

For example:



Earlier faults are displayed by indexing using the UP key. The fault log can be navigated using the UP/DOWN keys

6.6.2 Engineer's key parameter list

No.	Name	Description
EK1	Low Voltage Digital Input 1 (PA5–PA11)	Shows the state of each input. Where: 0 = OFF (Open circuit) 1 = ON
EK2	Low Voltage Digital Input 2 (PA6–PA11)	
EK3	Low Voltage Digital Input 3 (PA7–PA11)	
EK4	Low Voltage Digital Input 4 (PA8–PA11)	
EK5	Low Voltage 'HIGH' input (PA9–PA11)	
EK6	Low Voltage 'AUTO' input (P10–PA11)	
EK8	Profile 1 Select – Line Voltage (PE9)	
EK9	Profile 3 Select – Line Voltage (PE11)	
EK10	Profile 2 Select – Line Voltage (PE10)	
EK11	Profile 4 Select – Line Voltage (PE12)	